

Biology Chapter 7 — Metabolism

Your complete study pack — designed to actually make this fun.

Welcome! Metabolism sounds like something a doctor says before frowning at a blood report. But once you get past the jargon, it's one of the most mind-bending ideas in all of biology: **your body is a living chemistry set, running thousands of reactions every second, and every single one of those reactions is controlled by tiny protein machines called enzymes.**

Every time you move your finger, digest breakfast, or think about answering a question in class — that's metabolism happening. Right now. In your cells.

What's in this folder?

File	What it's for	When to use it
notes.md	Clean reference notes — all the facts, formulas, and definitions you need	Quick review before a test, or when writing a homework answer
explanations.md	Deep walkthroughs — the "why does this work?" explanations	When something in notes doesn't click. Read slowly.
stories-and-examples.md	Real-world examples and everyday stories that bring each topic to life	To understand WHY this matters outside the exam
history.md	The humans behind the science — Buchner, Krebs, Calvin, and the people who cracked the code of life	For curiosity. Also great exam-essay material.
analogies.md	Every topic reframed as something you already understand	When a concept feels abstract — these are your rescue rafts
flashcards.md	60+ Q&A cards organized by difficulty	Daily 10-minute review. Build memory fast.

The 7 topics in this chapter

1. **Metabolism and Its Types** — catabolism vs anabolism: tearing down vs building up
2. **Enzymes and Their Characteristics** — what they are, how they work, and why they're special
3. **Mechanism of Enzyme Action** — the lock-and-key story, active sites, ES complexes
4. **Factors Affecting Enzyme Activity** — temperature, pH, substrate concentration
5. **Enzyme Inhibition** — competitive vs non-competitive inhibition
6. **ATP — The Energy Currency of the Cell** — structure, the ADP cycle, high-energy bonds
7. **Photosynthesis and Respiration** — how cells make and spend their energy

A suggested study plan (1 week)

Day	What to do	Time
Mon	Read explanations.md topics 1-2 (metabolism + enzymes). Flashcards for those topics.	40 min
Tue	Read explanations.md topics 3-4 (enzyme mechanism + factors). Read the stories for enzymes.	50 min
Wed	Read explanations.md topic 5 (inhibition) and topic 6 (ATP). Go through analogies.md .	45 min
Thu	Read explanations.md topic 7 (photosynthesis + respiration). Draw the diagrams from memory.	50 min

Fri	Full <code>flashcards.md</code> session — cover all 60+ cards. Mark the ones you miss.	35 min
Sat	Read <code>stories-and-examples.md</code> cover to cover. Just enjoy it. No tests.	30 min
Sun	Quick review of <code>notes.md</code> . Focus on equations and the factors table.	20 min

One thing to remember before you start

The entire chapter rests on **one single idea**: life is chemistry, and chemistry needs energy to run.

Your body gets energy from food. Plants get energy from sunlight. But neither can use those raw energy sources directly — everything has to be converted into one universal "energy coin" called **ATP**. Enzymes are the machines that do the converting. Photosynthesis and respiration are the big pathways those machines run.

So when you're reading about enzyme active sites, or glycolysis, or the Calvin cycle — always ask yourself: *how does this connect back to making or spending ATP?*

That's the thread that ties this whole chapter together.

Good luck. You're going to love this one.